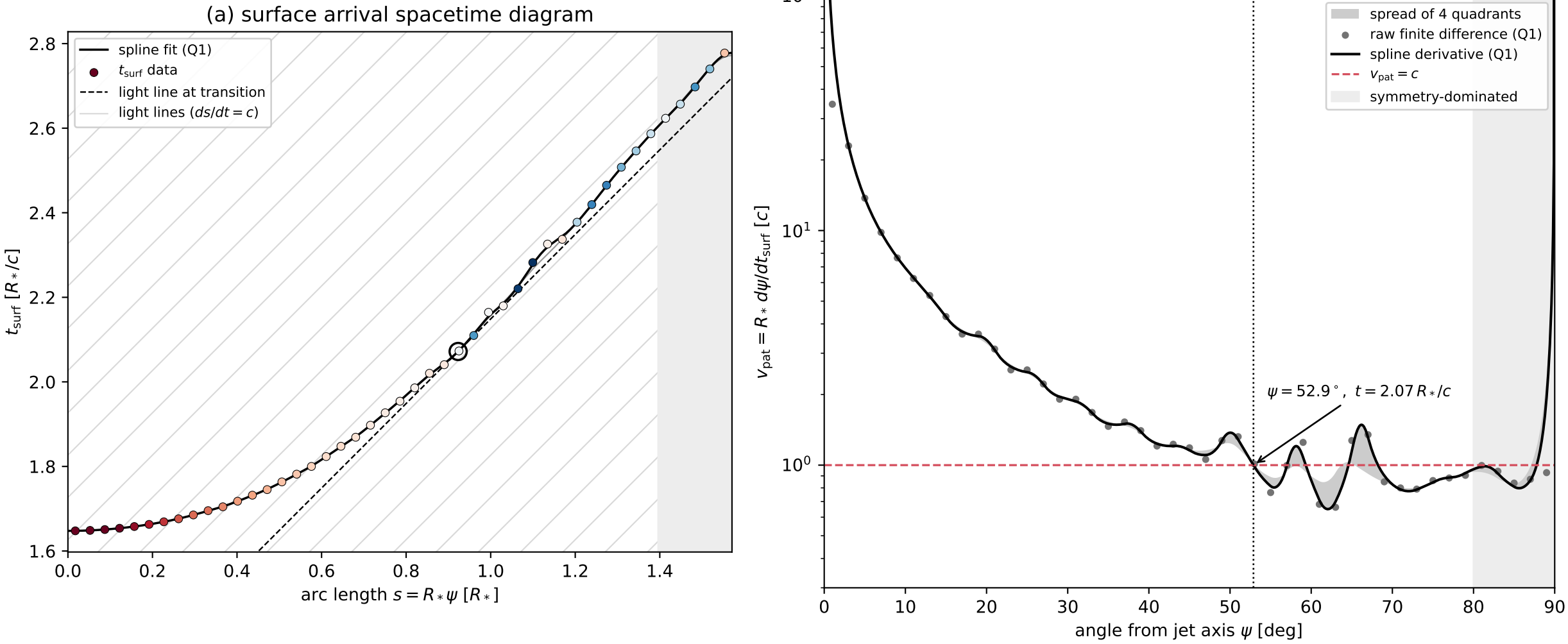
(a) surface arrival spacetime diagram
spline fit (Q1)
t_{surf} data
light line at transition
light lines ($ds/dt = c$)
t_{surf} $[R_*/c]$
arc length $s = R_*\psi$ $[R_*]$
(b) pattern speed along the surface
spread of 4 quadrants
raw finite difference (Q1)
spline derivative (Q1)
$v_{pat} = c$
symmetry-dominated
$\psi = 52.9°$, $t = 2.07\,R_*/c$
$v_{pat} = R_*\, d\psi/dt_{surf}$ $[c]$
angle from jet axis ψ [deg]